

TASK:

1. LED Data will increase from 0 to 16.
2. If button 1 is pressed, suspend the task
3. If button 2 is pressed, resume the task

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/* FreeRTOS includes. */
#include "FreeRTOS.h"
#include "task.h"
#include "xparameters.h"
#include "xgpio.h"
#include "xstatus.h"
#include "xil_printf.h"

/* Definitions */
#define INPUT_DEVICE_ID  XPAR_AXI_GPIO_0_DEVICE_ID
#define LED_DEVICE_ID    XPAR_AXI_GPIO_1_DEVICE_ID
#define LED_CHANNEL      1
#define BTN_CHANNEL      1
#define SW_CHANNEL       2
#define printf            xil_printf
#define DELAY_1_SECOND   1000UL

XGpio LEDInst, InputInst;

static void taskLED( void *taskLEDParams );
static void taskBTN( void *taskBTNParams );

static TaskHandle_t taskHandleLED;
static TaskHandle_t taskHandleBTN;

int main( void ) {

    int Status;

    Status = XGpio_Initialize(&LEDInst, LED_DEVICE_ID);
    if (Status != XST_SUCCESS) {
        return XST_FAILURE;
    }
    Status = XGpio_Initialize(&InputInst, INPUT_DEVICE_ID);
    if (Status != XST_SUCCESS) {
        return XST_FAILURE;
    }

    XGpio_SetDataDirection(&LEDInst, LED_CHANNEL, 0x00);

    XGpio_SetDataDirection(&InputInst, BTN_CHANNEL, 0xFF);

    XGpio_SetDataDirection(&InputInst, SW_CHANNEL, 0xFF);

    xTaskCreate(taskLED,
                (const char *) "LED Task",
                1000,
                NULL,
                tskIDLE_PRIORITY + 1,
                &taskHandleLED);

```

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    xTaskCreate(taskBTN, (const char *) "LED Task", 1000, NULL, tskIDLE_PRIORITY + 1,
&taskHandleBTN);

    /* Start the tasks. */
    vTaskStartScheduler();

    for( ;; );
}

static void taskLED( void *taskLEDParams ) {
    const TickType_t x1SecondInTicks = pdMS_TO_TICKS( DELAY_1_SECOND );

    /* Declare variables */
    int led_data = 0;
    int switch_data = 0;
    int output = 0;

    for( ;; ) {
        led_data = led_data+1;
        output = led_data;
        XGpio_DiscreteWrite(&LEDInst, LED_CHANNEL, output);
        vTaskDelay(x1SecondInTicks);
    }
}

static void taskBTN( void *taskBTNParams ) {
    int button_data = 0;

    for( ;; ) {
        button_data = XGpio_DiscreteRead(&InputInst, BTN_CHANNEL);

        if ((button_data & 0b0001) == 0b0001)
            vTaskSuspend(taskHandleLED);
        else if ((button_data & 0b0010) == 0b0010)
            vTaskResume(taskHandleLED);
    }
}

```